

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 4: Association Analysis

Ch 4 – Association Analysis – Numerical Problems & Solutions

5 methods · 14 PYQ instances · every itemset count recomputed from the printed transaction tables. **This is the heaviest numerical chapter in the syllabus.**

Methods

- **M1** Apriori: candidate generation, pruning, frequent itemsets
- **M2** Rule generation from frequent itemsets
- **M3** FP-tree construction and FP-Growth mining
- **M4** Pattern evaluation: lift and the correlation measures
- **M5** Subsequence enumeration

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-4 numerical below is built from

Move	What it says (where: problem, step; orange = key step)	Move	What it says (where: problem, step; orange = key step)
threshold to count	Turn min support into a COUNT first: $\lceil \text{fraction} \times N \rceil$. 1.3 1; 1.4/1.5 1; 1.6 1; 1.7 1; 1.8 1; 1.9 1; 1.10 1	node-links	Follow the item's node-links to every node carrying it. 3.1 1,2,3,4,5; 3.3 1,2,3,4,5
count and keep	Count each candidate over ALL transactions; keep those $\geq$ the min count. 1.1 1,2,3,4; 1.2 1,2,3; 1.3 2,3; 1.4/1.5 2,3,4,5; 1.6 2,3; 1.7 2,3,4; 1.8 2,3,4; 1.9 2,3; 1.10 2,3	prefix paths	Each node's whole root path, tagged with THAT node's count: the pattern base. 3.1 1,2,3,4,5; 3.3 1,2,3,4,5
join	Join two frequent $(k-1)$ -itemsets sharing their first $k-2$ items. 1.1 2,3,4,5; 1.2 2,3,4; 1.3 3,4; 1.4/1.5 3,4,5; 1.6 3; 1.7 3,4,5; 1.8 3,4,5; 1.9 3,4; 1.10 3,4	conditional tree	Add up the base per item, drop those below the min count, append the item. 3.1 1,2,3,4,5; 3.3 1,2,3,4,5
prune	Drop a candidate if any $(k-1)$ -subset is not frequent. No scan needed. 1.1 3; 1.4/1.5 4,5; 1.7 4; 1.8 4	lift	$P(A \cup B) / (P(A)P(B))$: below 1 is negative correlation, whatever the confidence. 4.1 3
stop	No candidates (or none survive): the run ends. 1.1 5; 1.2 4; 1.3 4; 1.4/1.5 5; 1.6 3; 1.7 5; 1.8 5; 1.9 4; 1.10 4	chi-square	$\sum (obs - exp)^2 / exp$ over the contingency table. 4.1 4
confidence	$\sigma(l) / \sigma(s)$: same numerator for every rule from l . 1.1 6; 1.4 1; 1.5 2; 1.6 4; 1.7 6; 1.8 6; 1.9 5; 1.10 5; M2 1; 3.1 1; 4.1 1,2	item splits	Enumerate how many items come from each element, then choose them. 5.1 1
threshold line	A rule is strong when its confidence is $\geq$ min_conf (equal counts). 1.1 6; 1.4 1; 1.5 2; 1.6 4; 1.7 6; 1.8 6; 1.9 5; 1.10 5; M2 1; 3.1 1	remove duplicates	The same sequence reached by two splits counts once. 5.1 2
F-list	Frequent items sorted by descending count, ties broken by a stated rule. –	keep element order	Subsets of each element, in the original element order, never merged. 5.2 1
reorder	Rewrite each transaction in F-list order, dropping infrequent items. –	common core	Remove one edge from each parent so the rest is identical: the core. 6.1 1
share prefix	A transaction that starts like an existing path increments its counts. 3.1 2,3,4,5; 3.2 2,3,5; 3.3 2,4,5,6; 3.4 2,4,5,6	attach edges	Add each parent's unique edge to the core at every legal vertex. 6.1 2,3
new branch	Where the prefix stops matching, a new child (or root child) is added. 3.1 1,2,4; 3.2 1,2,3,4; 3.3 1,3,4,5; 3.4 1,2,3,4,5	symmetry	Placements that map onto each other by a symmetry are one candidate. 6.1 2,3

Convert the threshold to a COUNT before you start. “min support 50%” on 6 transactions means a count of 3. Nearly every lost mark in this chapter comes from mixing percentages and counts half way through. Write $min_sup_count = N$ at the top and use only that.

Also: **support is over ALL transactions; confidence is over the antecedent's transactions only.**

M1. Apriori

Method

1. Scan once, count items $\rightarrow$ keep those $\geq \text{min_sup_count} \rightarrow L_1$.
 2. Join L_{k-1} with itself: combine 2 itemsets only if their first $k - 2$ items match.
 3. Prune: discard a candidate if *any* of its $(k - 1)$ -subsets is missing from L_{k-1} .
 4. Count survivors against D , keep those meeting the threshold $\rightarrow L_k$.
 5. Stop when L_k is empty.
- Show the prune step explicitly – it is where the method marks are.

Problem 1.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q3 (b) · [2+7]

What is Association Analysis? Explain with different use cases? Use the Apriori algorithm to find the frequent itemsets. Assume minimum support count is 4 and confidence is 80%.

TID	Items
T100	F, A, C, D, G, I, M, P, N
T101	A, B, C, D, F, L, M, O, P
T102	B, F, H, V, J, O, P
T103	B, C, K, S, A, V
T104	L, A, F, C, E, P, M, N, V
T105	I, B, A, P, S, M
T106	F, A, C, I, B, A, P

T106 genuinely prints "F, A, C, I, B, A, P" – **A is repeated; a transaction is a set**, so treat it as {F,A,C,I,B,P}.

Step 1: $C_1 \rightarrow L_1$

count all 18 distinct items over the 7 transactions, min count = 4

It.	Cnt	Keep	It.	Cnt	Keep	It.	Cnt	Keep	It.	Cnt	Keep
A	6	✓	F	5	✓	K	1	×	O	2	×
B	5	✓	G	1	×	L	2	×	P	6	✓
C	5	✓	H	1	×	M	4	✓	S	2	×
D	2	×	I	3	×	N	2	×	V	3	×
E	1	×	J	1	×						

$L_1 = \{A(6), B(5), C(5), F(5), M(4), P(6)\}$ – 6 items survive of 18.

Step 2: $C_2 \rightarrow L_2$

all $\binom{6}{2} = 15$ pairs from L_1 , counted against D

Pair	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep
AB	4	✓	AP	5	✓	BP	4	✓	CP	4	✓
AC	5	✓	BC	3	×	CF	4	✓	FM	3	×
AF	4	✓	BF	3	×	CM	3	×	FP	5	✓
AM	4	✓	BM	2	×				MP	4	✓

$L_2 = \{AB(4), AC(5), AF(4), AM(4), AP(5), BP(4), CF(4), CP(4), FP(5), MP(4)\}$

Step 3: $C_3 \rightarrow L_3$

join L_2 pairs sharing their first item, then **prune before counting**

Cand.	Subset test (all 2-subsets must be in L_2)	Transactions	Count	Keep
ABC	$BC \notin L_2$	pruned, never counted	—	×
ABF	$BF \notin L_2$	pruned, never counted	—	×
ABM	$BM \notin L_2$	pruned, never counted	—	×
ABP	$AB \checkmark AP \checkmark BP \checkmark \rightarrow$ survives pruning	T101, T105, T106	3	×
ACF	$AC \checkmark AF \checkmark CF \checkmark$	T100, T101, T104, T106	4	✓
ACM	$CM \notin L_2$	pruned, never counted	—	×
ACP	$AC \checkmark AP \checkmark CP \checkmark$	T100, T101, T104, T106	4	✓
AFM	$FM \notin L_2$	pruned, never counted	—	×
AFP	$AF \checkmark AP \checkmark FP \checkmark$	T100, T101, T104, T106	4	✓
AMP	$AM \checkmark AP \checkmark MP \checkmark$	T100, T101, T104, T105	4	✓
CFP	$CF \checkmark CP \checkmark FP \checkmark$	T100, T101, T104, T106	4	✓

$L_3 = \{\text{ACF}(4), \text{ACP}(4), \text{AFP}(4), \text{AMP}(4), \text{CFP}(4)\}$

- **Five candidates died at the prune step without a single database scan** – that saving is the whole point of the Apriori property, so show the column.
- ABP is the instructive one: it *passes* pruning and still fails on the count. **Pruning is sufficient to reject, never sufficient to accept.**

Step 4: $C_4 \rightarrow L_4$

join L_3 triples sharing their first two items

Cand.	Joined from	Subset test	Count	Keep
ACFP	ACF + ACP	$ACF \checkmark ACP \checkmark AFP \checkmark CFP \checkmark$	4 (T100, T101, T104, T106)	✓

$L_4 = \{\text{ACFP}(4)\}$; no two 4-itemsets remain to join $\rightarrow C_5 = \emptyset \rightarrow$ **stop**.

Frequent itemsets

Maximal frequent itemset = **{A, C, F, P} with support $4/7 = 57\%$.**

- **Check:** all $2^4 - 2 = 14$ proper subsets of ACFP do appear in $L_1 - L_3$ above. ✓

Step 5: the rules (the question gives confidence = 80 %, so this half is asked)

- From the maximal itemset $l = \{A, C, F, P\}$, $\sigma(l) = 4$. For every non-empty proper subset s , output $s \Rightarrow (l - s)$ with $conf = \frac{\sigma(l)}{\sigma(s)} = \frac{4}{\sigma(s)}$.
- $2^4 - 2 = 14$ candidate rules. **$conf \geq 0.80$ needs $\sigma(s) \leq 5$** , so read the answer straight off the antecedent's count.

Rule	$\sigma(s)$	Confidence
$C \Rightarrow AFP$	5	$4/5 = 80\% \checkmark$
$F \Rightarrow ACP$	5	$4/5 = 80\% \checkmark$
$AC \Rightarrow FP$	5	$4/5 = 80\% \checkmark$
$AP \Rightarrow CF$	5	$4/5 = 80\% \checkmark$
$FP \Rightarrow AC$	5	$4/5 = 80\% \checkmark$
$AF \Rightarrow CP$	4	$4/4 = 100\% \checkmark$
$CF \Rightarrow AP$	4	$4/4 = 100\% \checkmark$
$CP \Rightarrow AF$	4	$4/4 = 100\% \checkmark$
$ACF \Rightarrow P$	4	$4/4 = 100\% \checkmark$
$ACP \Rightarrow F$	4	$4/4 = 100\% \checkmark$
$AFP \Rightarrow C$	4	$4/4 = 100\% \checkmark$
$CFP \Rightarrow A$	4	$4/4 = 100\% \checkmark$
$A \Rightarrow CFP$	6	$4/6 = 66.7\% \times$
$P \Rightarrow ACF$	6	$4/6 = 66.7\% \times$

Reading the table

- **12 of the 14 rules are strong** at $\text{min_conf} = 80\%$.
- The only two failures are the ones with A or P alone on the left. Both occur in 6 of the 7 transactions, so they are **too common to predict anything**: a frequent antecedent divides the confidence down.
- **The general lesson**: confidence falls as the antecedent gets more frequent, and rises as it gets more specific. Longer left-hand sides almost always score higher.
- Seven rules here sit at **100%**. By §4.1 those are the *least* interesting of the twelve, because $\{A, F\}$ never appears without C and P anywhere in this database $\rightarrow$ nothing actionable. **Say so if the question asks you to comment**, do not just report the highest number.

Step by step: Apriori level by level, then the 14 rules



Problem 1.2 5

QUESTION AS PRINTED 2080 Baishakh · Q5 · [4+5]

Generally, we will be more interested in associated rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? Identify the candidate and large item sets of the following transaction table using Apriori algorithm with minimum support 2.

TID	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

Step 1: $C_1 \rightarrow L_1$

min count = 2

Item	In	Count	Keep
A	10, 30	2	✓
B	20, 30, 40	3	✓
C	10, 20, 30	3	✓
D	10	1	×
E	20, 30, 40	3	✓

 $L_1 = \{A(2), B(3), C(3), E(3)\}$ **Step 2:** $C_2 \rightarrow L_2$ all $\binom{4}{2} = 6$ pairs from L_1

Pair	In	Count	Keep
AB	30	1	×
AC	10, 30	2	✓
AE	30	1	×
BC	20, 30	2	✓
BE	20, 30, 40	3	✓
CE	20, 30	2	✓

 $L_2 = \{AC(2), BC(2), BE(3), CE(2)\}$ **Step 3:** $C_3 \rightarrow L_3$ join L_2 on a shared first item, prune, then count

Joined from	Candidate	Subset test	In	Count	Keep
BC + BE	BCE	BC ✓ BE ✓ CE ✓ → survives	20, 30	2	✓
AC + —	none	AC is the only pair starting with A, so it has nothing to join with	—	—	—

 $L_3 = \{BCE(2)\}$; C_4 needs two 3-itemsets sharing a prefix and there is only one → **stop**.**Final answer**

Frequent itemsets: A, B, C, E, AC, BC, BE, CE, BCE.

Step by step: Apriori level by level**1. $C_1 \rightarrow L_1$, min count 2**

count and keep

A 2 B 3 C 3 D 1 E 3

 L_1 : 4 kept of 5**2. $C_2 \rightarrow L_2$**

join count and keep

AB 1 AC 2 AE 1 BC 2 BE 3 CE 2

 L_2 : 4 kept of 6**3. $C_3 \rightarrow L_3$**

join count and keep

BCE 2

 L_3 : 1 kept of 1**4. $C_4 \rightarrow L_4$**

join stop

 $C_4 = \emptyset$: no two itemsets share a prefix.

KEY STEP

Problem 1.3 10

QUESTION AS PRINTED 2070 Chaitra · Q4 · [10]

Explain Apriori algorithm. Use Apriori to generate frequent item sets with support of 50% for the following transaction database.

TID	Items
1	ACD
2	BD
3	ABCE
4	BDF

4 transactions, 50% → min count = 2.

- L_1 : A(2), B(3), C(2), D(3) (E:1, F:1 dropped)
- C_2 : AB(1)× AC(2)✓ AD(1)× BC(1)× BD(2)✓ CD(1)×
- $L_2 = \{AC(2), BD(2)\}$
- C_3 : AC and BD share no common prefix → **no candidates**. Stop.

Final answer

Frequent itemsets: A, B, C, D, AC, BD.

Step by step: Apriori level by level

1. Min support as a count

threshold to count

4 transactions $\times$ 50% = 2

2. $C_1 \rightarrow L_1$, min count 2

count and keep

A 2 B 3 C 2 D 3 E 1 F 1

L_1 : 4 kept of 6

3. $C_2 \rightarrow L_2$

join count and keep

AB 1 AC 2 AD 1 BC 1 BD 2 CD 1

L_2 : 2 kept of 6

4. $C_3 \rightarrow L_3$

join stop

$C_3 = \emptyset$: no two itemsets share a prefix.

KEY STEP

Problems 1.4 and 1.5 5+3 8

QUESTION AS PRINTED 2076 Chaitra · Q7 · [5+3]

Consider the transaction data shown in the following table from a fast food restaurant.

Meal Item	List of Item IDs
Order:1	M1, M2, M5
Order:2	M2, M4
Order:3	M2, M3
Order:4	M1, M2, M4
Order:5	M1, M3
Order:6	M2, M3
Order:7	M1, M3
Order:8	M1, M2, M3, M5
Order:9	M1, M2, M3

There are 9 distinct transactions (Order: 1 – Order: 9) and each transaction involves between 2 and 4 meal items. There are a total of 5 meal items that are involved in the transactions. For simplicity, the meal items have been assigned short names (M1–M5). Assume that the minimum support is $2/9$ and the minimum confidence is $7/9$.

- Apply the Apriori algorithm to the dataset of transactions and identify all frequent k -itemsets.
- Find all strong association rules of the form: $X \wedge Y \rightarrow Z$ and note their confidence values.

QUESTION AS PRINTED 2074 Ashwin · Q5 · [8]

Identify the candidate, frequent item sets and association rules for the following transaction data using Apriori algorithm.

TID	ITEMS
1	M1, M2, M5
2	M2, M4
3	M2, M3
4	M1, M2, M4
5	M1, M3
6	M2, M3
7	M1, M3
8	M1, M2, M3, M5
9	M1, M2, M3

Take minimum support = 20%, minimum confidence 80%

Identical 9-transaction table, and both thresholds work out to a count of 2, so the frequent itemsets are the same; only the rule filter differs.

Frequent itemsets

(9 transactions, min count 2)

Step 1: $C_1 \rightarrow L_1$

Item	In	Count	Keep
M1	T1,4,5,7,8,9	6	✓
M2	T1,2,3,4,6,8,9	7	✓
M3	T3,5,6,7,8,9	6	✓
M4	T2,4	2	✓
M5	T1,8	2	✓

Nothing is dropped — all 5 items clear a count of 2.

Step 2: $C_2 \rightarrow L_2$

all $\binom{5}{2} = 10$ pairs

Pair	In	Count	Keep
M1M2	T1, T4, T8, T9	4	✓
M1M3	T5, T7, T8, T9	4	✓
M1M4	T4	1	×
M1M5	T1, T8	2	✓
M2M3	T3, T6, T8, T9	4	✓
M2M4	T2, T4	2	✓
M2M5	T1, T8	2	✓
M3M4	—	0	×
M3M5	T8	1	×
M4M5	—	0	×

$L_2 = \{M1M2(4), M1M3(4), M1M5(2), M2M3(4), M2M4(2), M2M5(2)\}$

Step 3: $C_3 \rightarrow L_3$

join two L_2 pairs **only when both are in L_2 and they share their first item**

Joined from	Candidate	Subset test	In	Count	Keep
M1M2 + M1M3	M1M2M3	M1M2 ✓ M1M3 ✓ M2M3 ✓	T8, T9	2	✓
M1M2 + M1M5	M1M2M5	M1M2 ✓ M1M5 ✓ M2M5 ✓	T1, T8	2	✓
M1M3 + M1M5	M1M3M5	M3M5 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M3 + M2M4	M2M3M4	M3M4 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M3 + M2M5	M2M3M5	M3M5 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M4 + M2M5	M2M4M5	M4M5 $\notin L_2 \rightarrow$ pruned	—	—	×

$L_3 = \{M1M2M3(2), M1M2M5(2)\}$

- Note what is not in the table:** M1M2M4 never appears, because generating it would need M1M4, and M1M4 failed at step 2. **A candidate is only ever built from two surviving parents** — it is not built and then pruned.

Step 4: $C_4 \rightarrow L_4$

Joined from	Candidate	Subset test	Count	Keep
M1M2M3 + M1M2M5	M1M2M3M5	M1M3M5 $\notin L_3 \rightarrow$ pruned	—	×

$L_4 = \emptyset \rightarrow$ **stop**.

Step by step: Apriori level by level (both papers)**1. Min support as a count**

threshold to count

$2/9$ of $9 = 2$; 20% of $9 = 1.8 \rightarrow 2$

2. $C_1 \rightarrow L_1$, min count 2

count and keep

M1 6 M2 7 M3 6 M4 2 M5 2

L_1 : 5 kept of 5

3. $C_2 \rightarrow L_2$

join count and keep

M1M2 4 M1M3 4 M1M4 1 M1M5 2 M2M3 4
M2M4 2 M2M5 2 M3M4 0 M3M5 1 M4M5 0

L_2 : 6 kept of 10

4. $C_3 \rightarrow L_3$

join prune count and keep

M1M2M3 2 M1M2M5 2 M1M3M5
M2M3M4 M2M3M5 M2M4M5

pruned (dashed): M1M3M5 ($M3M5 \notin L_2$), M2M3M4 ($M3M4 \notin L_2$), M2M3M5 ($M3M5 \notin L_2$), M2M4M5 ($M4M5 \notin L_2$)
 C_3 : 2 kept of 6

KEY STEP

5. $C_4 \rightarrow L_4$

join prune count and keep stop

M1M2M3M5

pruned (dashed): M1M2M3M5 ($M1M3M5 \notin L_3$)

$L_4 = \emptyset$: nothing survives

The one idea that decides every rule below

- Every rule out of an itemset l has the *same* numerator: $conf(s \Rightarrow l - s) = \sigma(l)/\sigma(s)$. **Only the antecedent's count moves.**

- So a rule is strong exactly when **the antecedent is nearly as rare as the whole itemset** → when the antecedent hardly ever appears without the consequent.
- Here $\sigma(M4) = \sigma(M5) = 2$, and both *always* appear with their partners. **Expect every strong rule to have M4 or M5 on the left**, and check that prediction against the tables. Nothing else can win, because M1, M2 and M3 each occur 6–7 times while no itemset containing them has support above 4.

1.4 Rules of the form $X \wedge Y \Rightarrow Z$, $\min \text{conf} = 7/9 = 0.778$

- That form means a **2-item antecedent and a 1-item consequent**, so only the **3-itemsets** in L_3 can produce one. Two of them, three rules each → **6 candidates, and they are all of them**.

From	Rule	$\sigma(XY)$	$\text{Conf} = 2/\sigma(XY)$
M1M2M3 $\sigma = 2$	M1 $\wedge$ M2 $\Rightarrow$ M3	4	0.50 $\times$
	M1 $\wedge$ M3 $\Rightarrow$ M2	4	0.50 $\times$
	M2 $\wedge$ M3 $\Rightarrow$ M1	4	0.50 $\times$
M1M2M5 $\sigma = 2$	M1 $\wedge$ M2 $\Rightarrow$ M5	4	0.50 $\times$
	M1$\wedge$M5$\Rightarrow$M2	2	1.00 $\checkmark$
	M2$\wedge$M5$\Rightarrow$M1	2	1.00 $\checkmark$

Answer to 1.4

- **2 strong rules, both at 100 % confidence:**
M1 $\wedge$ M5 $\Rightarrow$ M2 and M2 $\wedge$ M5 $\Rightarrow$ M1.
- Every rule out of M1M2M3 fails, and all three fail *identically* at 0.50, because all three of its 2-subsets happen to have $\sigma = 4$.
- The two winners are exactly the ones predicted above: **M5 is in the antecedent**, and M5 occurs only in T1 and T8, both of which also hold M1 and M2.

1.5 All association rules, $\min \text{conf} = 0.80$

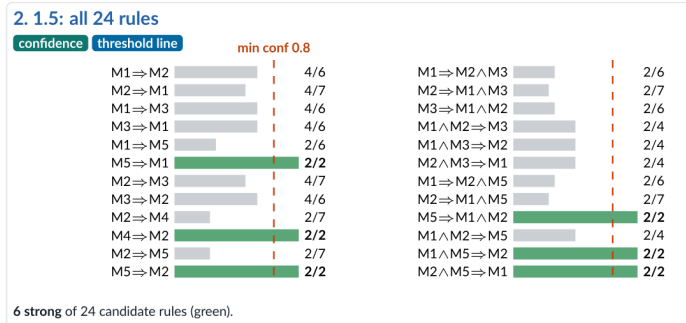
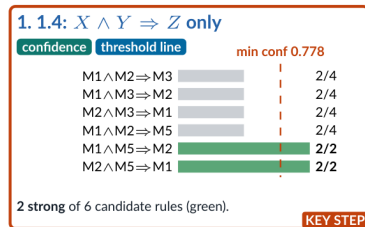
- Now every frequent itemset with $|I| \geq 2$ contributes: six 2-itemsets (2 rules each) and two 3-itemsets ($2^3 - 2 = 6$ rules each) → **12 + 12 = 24 candidate rules**. All 24 are listed, none is skipped.

From	Rule	$\sigma(s)$	Conf	From	Rule	$\sigma(s)$	Conf
M1M2 (4)	M1 $\Rightarrow$ M2	6	0.67 $\times$	M2M5 (2)	M2 $\Rightarrow$ M5	7	0.29 $\times$
	M2 $\Rightarrow$ M1	7	0.57 $\times$		M5$\Rightarrow$M2	2	1.00 $\checkmark$
M1M3 (4)	M1 $\Rightarrow$ M3	6	0.67 $\times$	M1M2M3 (2)	M1 $\Rightarrow$ M2M3	6	0.33 $\times$
	M3 $\Rightarrow$ M1	6	0.67 $\times$		M2 $\Rightarrow$ M1M3	7	0.29 $\times$
M1M5 (2)	M1 $\Rightarrow$ M5	6	0.33 $\times$		M3 $\Rightarrow$ M1M2	6	0.33 $\times$
	M5$\Rightarrow$M1	2	1.00 $\checkmark$		M1 $\wedge$ M2 $\Rightarrow$ M3	4	0.50 $\times$
M2M3 (4)	M2 $\Rightarrow$ M3	7	0.57 $\times$		M1 $\wedge$ M3 $\Rightarrow$ M2	4	0.50 $\times$
	M3 $\Rightarrow$ M2	6	0.67 $\times$	M1M2M5 (2)	M2 $\wedge$ M3 $\Rightarrow$ M1	4	0.50 $\times$
M2M4 (2)	M2 $\Rightarrow$ M4	7	0.29 $\times$		M1 $\Rightarrow$ M2M5	6	0.33 $\times$
	M4$\Rightarrow$M2	2	1.00 $\checkmark$		M2 $\Rightarrow$ M1M5	7	0.29 $\times$
					M5$\Rightarrow$M1$\wedge$M2	2	1.00 $\checkmark$
					M1 $\wedge$ M2 $\Rightarrow$ M5	4	0.50 $\times$
					M1$\wedge$M5$\Rightarrow$M2	2	1.00 $\checkmark$
					M2$\wedge$M5$\Rightarrow$M1	2	1.00 $\checkmark$

Answer to 1.5

- **6 strong rules, every one of them at 100 % confidence:**
M5 $\Rightarrow$ M1 M5 $\Rightarrow$ M2 M4 $\Rightarrow$ M2 M5 $\Rightarrow$ M1 $\wedge$ M2 M1 $\wedge$ M5 $\Rightarrow$ M2 M2 $\wedge$ M5 $\Rightarrow$ M1
- The other 18 fail, and **the prediction held**: every survivor has M4 or M5 on the left. M1, M2 and M3 are too frequent to be confident antecedents here, the best any of them manages is 0.67.
- **Cross-check with 1.4**: the two rules found there must reappear here, since 1.4's threshold (0.778) is the tighter of the two. They do, on the last two rows. $\checkmark$
- **Worth a closing sentence**: all six sit at 100 %, which by §4.1 makes them *reliable but not surprising* → M4 and M5 are simply never ordered alone. Confidence alone cannot see that; lift would.

Step by step: the rules for each paper's threshold



Problem 1.6 8

QUESTION AS PRINTED 2079 Bhadra · Q5 · [8]

Derive association rule for the following market basket transactions.

Minimum support = 50% Minimum confidence = 80%

Transaction ID	Item Set
1	A,B
2	A,D
3	A,C
4	B,E
5	B,D,E
6	A,E,C

This one has a surprise. 6 transactions, 50% $\rightarrow$ min count = 3.Step 1: $C_1 \rightarrow L_1$

Item	In	Count	Keep
A	1, 2, 3, 6	4	✓
B	1, 4, 5	3	✓
C	3, 6	2	×
D	2, 5	2	×
E	4, 5, 6	3	✓

 $L_1 = \{A(4), B(3), E(3)\}$

Final answer

- No frequent 2-itemset exists, so no association rule can be derived at 50% support. Say this explicitly – it is the correct answer, not a dead end.
- And it holds for every confidence threshold, including 0%: a rule needs a 2-itemset to split, and there is none. Lowering min_conf cannot rescue it, only lowering min_sup can (§4.2).
- Show you understand why: the data is spread thin, every pair occurs at most twice.
- If min support is relaxed to 33.3% (count 2), the picture changes:

Step 2: $C_2 \rightarrow L_2$ the 3 pairs from L_1

Pair	In	Count	Keep
AB	1	1	×
AE	6	1	×
BE	4, 5	2	×

 $L_2 = \emptyset$ – every pair falls short.

New L_2	Count	Rule	Confidence	Strong at 80 %?
AC	2	$A \Rightarrow C$	$2/4 = 0.50$	×
		$C \Rightarrow A$	$2/2 = 1.00$	✓
BE	2	$B \Rightarrow E$	$2/3 = 0.67$	×
		$E \Rightarrow B$	$2/3 = 0.67$	×

One rule survives: $C \Rightarrow A$ at 100%. Worth adding as a closing remark. verified/added

Step by step: nothing frequent at 50%, then the relaxed run

1. Min support as a count

threshold to count

6 transactions $\times$ 50% = 32. $C_1 \rightarrow L_1$, min count 3

count and keep

A 4 B 3 C 2 D 2 E 3

 L_1 : 3 kept of 53. $C_2 \rightarrow L_2$

join count and keep stop

AB 1 AE 1 BE 2

 $L_2 = \emptyset$: nothing survives

KEY STEP

4. Relaxed to count 2: the rules

confidence

threshold line

min conf 0.8

$A \Rightarrow C$	2/4
$C \Rightarrow A$	2/2
$B \Rightarrow E$	2/3
$E \Rightarrow B$	2/3

1 strong of 4 candidate rules (green).

Problem 1.7 2+8

QUESTION AS PRINTED 2082 Bhadra • Q4 • [2+8]

Explain the term support and confidence in association analysis. Derive association rule for following transaction considering minimum support = 60% and confidence = 80% using Apriori Algorithm.

TID Transaction

T1	A, B, C, D, E, F
T2	B, C, D, E, F, G
T3	A, D, E, H
T4	A, D, F, I, J
T5	B, D, E, K

5 transactions, 60% $\rightarrow$ min count = 3.Step 1: $C_1 \rightarrow L_1$

11 distinct items

Item	In	Cnt	Keep
A	T1, T3, T4	3	✓
B	T1, T2, T5	3	✓
C	T1, T2	2	×
D	T1-T5	5	✓
E	T1, T2, T3, T5	4	✓
F	T1, T2, T4	3	✓
G	T2	1	×
H	T3	1	×
I	T4	1	×
J	T4	1	×
K	T5	1	×

 $L_1 = \{A(3), B(3), D(5), E(4), F(3)\}$ Step 2: $C_2 \rightarrow L_2$ all $\binom{5}{2} = 10$ pairs

Pair	In	Cnt	Keep
AB	T1	1	×
AD	T1, T3, T4	3	✓
AE	T1, T3	2	×
AF	T1, T4	2	×
BD	T1, T2, T5	3	✓
BE	T1, T2, T5	3	✓
BF	T1, T2	2	×
DE	T1, T2, T3, T5	4	✓
DF	T1, T2, T4	3	✓
EF	T1, T2	2	×

 $L_2 = \{AD(3), BD(3), BE(3), DE(4), DF(3)\}$ Step 3: $C_3 \rightarrow L_3$

Joined from	Cand.	Subset test	In	Cnt	Keep
BD + BE	BDE	BD ✓ BE ✓ DE ✓	T1, T2, T5	3	✓
AD + —	none	AD is the only pair starting with A	—	—	—
DE + DF	DEF	EF $\notin L_2 \rightarrow$ pruned	—	—	×

 $L_3 = \{BDE(3)\}; C_4 = \emptyset \rightarrow$ stop.

Step 4: the rules

 $5 \times 2 + (2^3 - 2) = 16$ candidates, min conf 80%

From	Rule	$\sigma(s)$	Conf	From	Rule	$\sigma(s)$	Conf
AD (3)	$A \Rightarrow D$	3	1.00 ✓	DF (3)	$D \Rightarrow F$	5	0.60 ×
	$D \Rightarrow A$	5	0.60 ×		$F \Rightarrow D$	3	1.00 ✓
BD (3)	$B \Rightarrow D$	3	1.00 ✓	BDE (3)	$B \Rightarrow DE$	3	1.00 ✓
	$D \Rightarrow B$	5	0.60 ×		$D \Rightarrow BE$	5	0.60 ×
BE (3)	$B \Rightarrow E$	3	1.00 ✓		$E \Rightarrow BD$	4	0.75 ×
	$E \Rightarrow B$	4	0.75 ×		$BD \Rightarrow E$	3	1.00 ✓
DE (4)	$D \Rightarrow E$	5	0.80 ✓		$BE \Rightarrow D$	3	1.00 ✓
	$E \Rightarrow D$	4	1.00 ✓		$DE \Rightarrow B$	4	0.75 ×

- **9 strong rules of 16.** $D \Rightarrow E$ scrapes in at exactly 0.80 – **$\geq$ the threshold counts as passing**, so keep it.
- Every failure has D or E alone on the left. D is in *all five* transactions, so it predicts nothing (§4.1); it is the consequent of four winning rules instead.

Step by step: Apriori level by level, then the 16 rules

1. Min support as a count

threshold to count

5 transactions $\times$ 60% = 3

2. $C_1 \rightarrow L_1$, min count 3

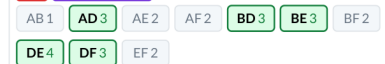
count and keep



L_1 : 5 kept of 11

3. $C_2 \rightarrow L_2$

join count and keep



L_2 : 5 kept of 10

4. $C_3 \rightarrow L_3$

join prune count and keep



pruned (dashed): DEF ($EF \notin L_2$)

L_3 : 1 kept of 2

KEY STEP

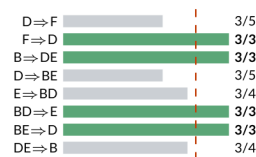
5. $C_4 \rightarrow L_4$

join stop

$C_4 = \emptyset$: no two itemsets share a prefix.

6. Every rule against min conf

confidence threshold line min conf 0.8



9 strong of 16 candidate rules (green).

Problem 1.8 2+6

QUESTION AS PRINTED 2081 Bhadra · Q7 · [2+6]

For the transaction given below, consider Confidence = 65% and minimum support = 50%. Identify frequent itemsets with possible association rules using Apriori Algorithm.

TID	Items
1	MILK, BREAD, CAKE
2	BUTTER, BREAD, EGG
3	MILK, BUTTER, BREAD, EGG
4	BUTTER, EGG

4 transactions, 50% $\rightarrow$ min count = 2. Note the unusually low confidence bar, 65%, which is why almost everything survives.

Step 1: $C_1 \rightarrow L_1$

Item	In	Cnt	Keep
MILK	1, 3	2	✓
BREAD	1, 2, 3	3	✓
CAKE	1	1	×
BUTTER	2, 3, 4	3	✓
EGG	2, 3, 4	3	✓

 $L_1 = \{\text{MILK}(2), \text{BREAD}(3), \text{BUTTER}(3), \text{EGG}(3)\}$
Step 2: $C_2 \rightarrow L_2$

6 pairs

Pair	In	Cnt	Keep
BREAD, BUTTER	2, 3	2	✓
BREAD, EGG	2, 3	2	✓
BREAD, MILK	1, 3	2	✓
BUTTER, EGG	2, 3, 4	3	✓
BUTTER, MILK	3	1	×
EGG, MILK	3	1	×

Step 3: $C_3 \rightarrow L_3$

Joined from	Candidate	Subset test	Cnt	Keep
{BR,BU} + {BR,EG}	BREAD, BUTTER, EGG	all three 2-subsets in L_2 ✓	2	✓
{BR,BU} + {BR,MI}	BREAD, BUTTER, MILK	{BU,MI} $\notin L_2 \rightarrow$ pruned	—	×
{BR,EG} + {BR,MI}	BREAD, EGG, MILK	{EG,MI} $\notin L_2 \rightarrow$ pruned	—	×

 $L_3 = \{\{\text{BREAD, BUTTER, EGG}\}(2)\}$, $C_4 = \emptyset \rightarrow$ stop.
Step 4: the rules
 $4 \times 2 + 6 = 14$ candidates, min conf 65 %

Rule	$\sigma(s)$	Conf	Rule	$\sigma(s)$	Conf
BREAD $\Rightarrow$ BUTTER	3	0.667 ✓	BUTTER $\Rightarrow$ EGG	3	1.00 ✓
BUTTER $\Rightarrow$ BREAD	3	0.667 ✓	EGG $\Rightarrow$ BUTTER	3	1.00 ✓
BREAD $\Rightarrow$ EGG	3	0.667 ✓	BREAD $\Rightarrow$ BU,EG	3	0.667 ✓
EGG $\Rightarrow$ BREAD	3	0.667 ✓	BUTTER $\Rightarrow$ BR,EG	3	0.667 ✓
BREAD $\Rightarrow$ MILK	3	0.667 ✓	EGG $\Rightarrow$ BR,BU	3	0.667 ✓
MILK $\Rightarrow$ BREAD	2	1.00 ✓	BR,BU $\Rightarrow$ EGG	2	1.00 ✓
			BR,EG $\Rightarrow$ BUTTER	2	1.00 ✓
			BU,EG $\Rightarrow$ BREAD	3	0.667 ✓

- **All 14 rules are strong.** That is the point of the 65 % threshold: the lowest confidence anywhere in this database is $2/3 = 0.667$, which clears it.
- **Say so explicitly rather than listing 14 rules and stopping** — “every candidate rule passes, because min_conf sits below the minimum achievable confidence here” is the observation worth a mark. Raise the bar to 70 % and only the five 100 % rules survive.

Step by step: Apriori level by level, then the 14 rules

Problem 1.9 3+7

QUESTION AS PRINTED 2073 Chaitra · Q5 · [3+7]

Explain Apriori algorithm in market basket analysis? Derive association rule from the following market basket transactions with 50% of minimum support and confidence respectively.

Transaction	Itemsets
1	A, B, C
2	A, C
3	A, D
4	B, E, F

"50% of minimum support and confidence respectively" means **both** thresholds are 50%: min count = 2, min conf = 0.50.

Step 1: $C_1 \rightarrow L_1$

Item	In	Cnt	Keep
A	1, 2, 3	3	✓
B	1, 4	2	✓
C	1, 2	2	✓
D	3	1	×
E	4	1	×
F	4	1	×

$L_1 = \{A(3), B(2), C(2)\}$

Step 2: $C_2 \rightarrow L_2$

Pair	In	Cnt	Keep
AB	1	1	×
AC	1, 2	2	✓
BC	1	1	×

$L_2 = \{AC(2)\}$; only one pair survives, so $C_3 = \emptyset \rightarrow$ stop.

Step 3: the rules

only $\{A, C\}$ can produce any, so $2^2 - 2 = 2$ candidates

Rule	$\sigma(s)$	Confidence	Strong at 50%?
$A \Rightarrow C$	3	$2/3 = 0.667$	✓
$C \Rightarrow A$	2	$2/2 = 1.000$	✓

- **Both are strong:** $A \Rightarrow C$ (67%) and $C \Rightarrow A$ (100%).
- **Transaction 4 contributes nothing.** $\{B, E, F\}$ shares no item pattern with the rest, which is why B survives as a 1-itemset but never appears in a rule. Worth a sentence.

Step by step: Apriori level by level, then the two rules

1. Min support as a count

threshold to count

50% of 4 = 2; min conf = 0.50

2. $C_1 \rightarrow L_1$, min count 2

count and keep

A3 B2 C2 D1 E1 F1

L_1 : 3 kept of 6

4. $C_3 \rightarrow L_3$

join stop

$C_3 = \emptyset$; no two itemsets share a prefix.

5. Every rule against min conf

confidence threshold line min conf 0.5

A $\Rightarrow$ C 2/3
C $\Rightarrow$ A 2/2

2 strong of 2 candidate rules (green).

3. $C_2 \rightarrow L_2$

join count and keep

AB1 AC2 BC1

L_2 : 1 kept of 3

KEY STEP

Problem 1.10 5+3

QUESTION AS PRINTED 2079 Baishakh · Q5 · [5+3]

a) Write Apriori algorithm and using the algorithm find all the frequent itemset for the following database. ($\text{min_sup} = 20\%$).

TID	A1	A2	A3	A4	A5	A6	A7	A8	A9
T2	0	1	0	1	0	0	0	1	0
T3	0	0	0	1	1	0	1	0	0
T4	0	0	1	0	0	0	0	0	0
T5	0	0	0	0	1	1	1	0	0
T6	0	1	1	1	0	0	0	0	0
T7	0	1	0	0	0	1	1	0	1
T8	0	0	0	0	1	0	0	0	0

b) Is it possible to generate any rules out of the frequent items, considering any value of confidence threshold?

The table really does start at T2, has seven rows, and its 8th column really is headed A8 (the scan clips the 8). 20% of 7 transactions = 1.4, and a count cannot be fractional, so $\text{min count} = 2$. Column A1 is 0 in every row — it can never be frequent, so drop it at once and say why.

(a) Step 1: $C_1 \rightarrow L_1$ Step 2: $C_2 \rightarrow L_2$ $\binom{6}{2} = 15$ pairs

Item	In	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep
A1	— (all zero)	0	×	A2,A3	1	×	A4,A5	1	×
A2	T2, T6, T7	3	✓	A2,A4	2	✓	A4,A6	0	×
A3	T4, T6	2	✓	A2,A5	0	×	A4,A7	1	×
A4	T2, T3, T6	3	✓	A2,A6	1	×	A5,A6	1	×
A5	T3, T5, T8	3	✓	A2,A7	1	×	A5,A7	2	✓
A6	T5, T7	2	✓	A3,A4	1	×	A6,A7	2	✓
A7	T3, T5, T7	3	✓	A3,A5	0	×			
A8	T2	1	×	A3,A6	0	×			
A9	T7	1	×	A3,A7	0	×			

Step 3: C_3

- $L_2 = \{A2A4(2), A5A7(2), A6A7(2)\}$.
- To build a 3-itemset two pairs must **share their first item**. A2A4, A5A7 and A6A7 all start with a different item, so **no candidate can be formed at all** $\rightarrow C_3 = \emptyset \rightarrow \text{stop}$.
- Frequent itemsets: A2, A3, A4, A5, A6, A7, {A2,A4}, {A5,A7}, {A6,A7}.

(b) Can any rule be generated, at any confidence threshold?

- Yes**. The question is a trap only if you stop at L_1 : a rule needs a frequent itemset of size ≥ 2 to split into antecedent and consequent, and here **three 2-itemsets survive**, so rules exist.
- Each 2-itemset yields $2^2 - 2 = 2$ rules, so **6 rules in total**:

Rule	$\sigma(s)$	Confidence	Comment
A2 $\Rightarrow$ A4	3	$2/3 = 0.667$	
A4 $\Rightarrow$ A2	3	$2/3 = 0.667$	
A5 $\Rightarrow$ A7	3	$2/3 = 0.667$	
A7 $\Rightarrow$ A5	3	$2/3 = 0.667$	
A6 $\Rightarrow$ A7	2	$2/2 = 1.000$	A6 never occurs without A7
A7 $\Rightarrow$ A6	3	$2/3 = 0.667$	

- So the honest answer is: **rules exist for any threshold up to 0.667**, and above that only **A6 $\Rightarrow$ A7** survives. Above 1.0 nothing does.
- The general principle** (§4.2): rules exist iff $L_2 \neq \emptyset$. Check L_2 , not the confidence value — if L_2 were empty no threshold, not even 0 %, could produce a rule. Compare Problem 1.6, where exactly that happens.

Step by step: Apriori level by level, then (b) the six rules

1. Min support as a count

threshold to count

20% of 7 = 1.4, a count cannot be fractional → 2

2. $C_1 \rightarrow L_1$, min count 2

count and keep

A2 3 A3 2 A4 3 A5 3 A6 2 A7 3 A8 1
A9 1

L_1 : 6 kept of 8

3. $C_2 \rightarrow L_2$

join count and keep

A2,A3 1 **A2,A4 2** A2,A5 0 A2,A6 1
A2,A7 1 A3,A4 1 A3,A5 0 A3,A6 0
A3,A7 0 A4,A5 1 A4,A6 0 A4,A7 1
A5,A6 1 **A5,A7 2** **A6,A7 2**

L_2 : 3 kept of 15

4. $C_3 \rightarrow L_3$

join stop

$C_3 = \emptyset$: no two itemsets share a prefix.

KEY STEP

5. (b) Every rule, at any threshold

confidence threshold line

A2 $\Rightarrow$ A4 2/3
A4 $\Rightarrow$ A2 2/3
A5 $\Rightarrow$ A7 2/3
A7 $\Rightarrow$ A5 2/3
A6 $\Rightarrow$ A7 2/2
A7 $\Rightarrow$ A6 2/3

6 strong of 6 candidate rules (green).

M2. Rule generation from frequent itemsets

Method

- For each frequent itemset l with $|l| \geq 2$, take every non-empty proper subset s :

$$s \Rightarrow (l - s) \quad \text{with} \quad \text{conf} = \frac{\text{sup_count}(l)}{\text{sup_count}(s)}$$

- Keep it if $\text{conf} \geq \text{min_conf}$. A k -itemset gives $2^k - 2$ candidate rules.
- The support of every rule from l is the same** ($= \text{sup}(l)$); only confidence varies.
- Shortcut:** if $s \Rightarrow (l - s)$ fails, every rule whose antecedent is a subset of s also fails. Prune downward.

Worked example

PROBLEM not a PYQ · the rule half of 2081 Baishakh Q3 (b) **verified/added**

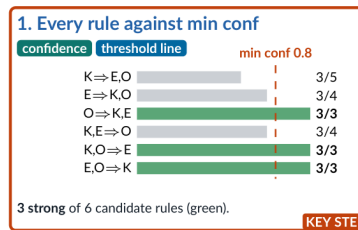
The frequent itemset $l = \{K, E, O\}$ has support count 3 in the 5-transaction database of Problem 3.1, where $\text{sup}(K) = 5$, $\text{sup}(E) = 4$, $\text{sup}(O) = 3$, $\text{sup}(K,E) = 4$, $\text{sup}(K,O) = 3$ and $\text{sup}(E,O) = 3$. List every association rule that l can generate and its confidence, and say which are strong at a minimum confidence of 80%.

From $l = \{K,E,O\}$ with $\text{sup} = 3$ (see Problem 3.1), $|l| = 3 \rightarrow 6$ candidate rules:

Rule	Confidence	Value	$\geq 80\%$?
$K \Rightarrow E,O$	3/5	0.60	×
$E \Rightarrow K,O$	3/4	0.75	×
$O \Rightarrow K,E$	3/3	1.00	✓
$K,E \Rightarrow O$	3/4	0.75	×
$K,O \Rightarrow E$	3/3	1.00	✓
$E,O \Rightarrow K$	3/3	1.00	✓

- Every rule with **O** in the antecedent succeeds, because O never appears without K and E.

Step by step: the six rules from K, E, O



M3. FP-tree construction and FP-Growth mining

Method

- Scan 1:** count items, drop infrequent ones, sort the rest by descending support → **F-list**. **State your tie-break rule.**
- Scan 2:** rewrite each transaction in F-list order, insert as a path, sharing prefixes and incrementing counts. Build the header table with node-links.
- Mine bottom-up** (least frequent item first): conditional pattern base → conditional FP-tree → recurse.

Problem 3.1 2+7

QUESTION AS PRINTED 2081 Baishakh · Q3 (b) · [2+7]

What is limitation of Apriori algorithm compared to FP-growth? A database has 5 transactions, given in table below. Let min support = 60% and min confidence = 80%.

TID	Item_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

- Find all frequent itemsets using FP-growth.
- List all of the strong association rules (with support and confidence).

- T500 prints O twice** — a transaction is a set, so it is {C,O,K,I,E}.
- 5 transactions, 60% → **min count = 3**.

Step 1: counts and F-list

- K:5, E:4, M:3, O:3, Y:3 ✓ N:2, C:2, D:1, A:1, U:1, I:1 ✗
- F-list = K(5), E(4), M(3), O(3), Y(3)** (ties at 3 broken alphabetically)

Step 2: reordered transactions

- T100 → K, E, M, O, Y
- T200 → K, E, O, Y
- T300 → K, E, M
- T400 → K, M, Y
- T500 → K, E, O

Step 3: the FP-tree

Drawn below, after every transaction; the last panel is the finished tree.

- Check:** counts on each level sum correctly — K:5 = E:4 + M:1. ✓

The node-link (header) table

- One entry per F-list item, pointing at every node carrying that label:

K → K:5

E → E:4

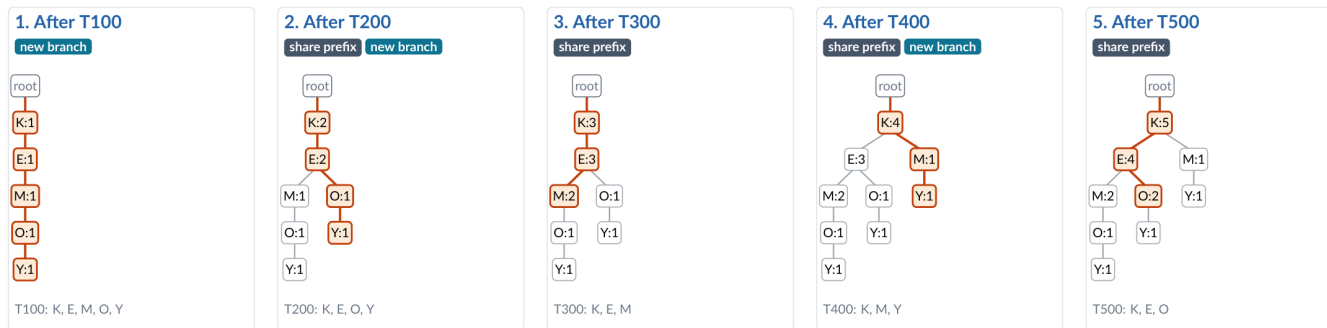
M → M:2 (under E), M:1 (under K)

O → O:1 (under M), O:2 (under E)

Y → Y:1 × 3 (three different branches)

- This table is what makes mining possible** — it is how you find all occurrences of an item without walking the tree.

Step by step: the FP-tree after each transaction



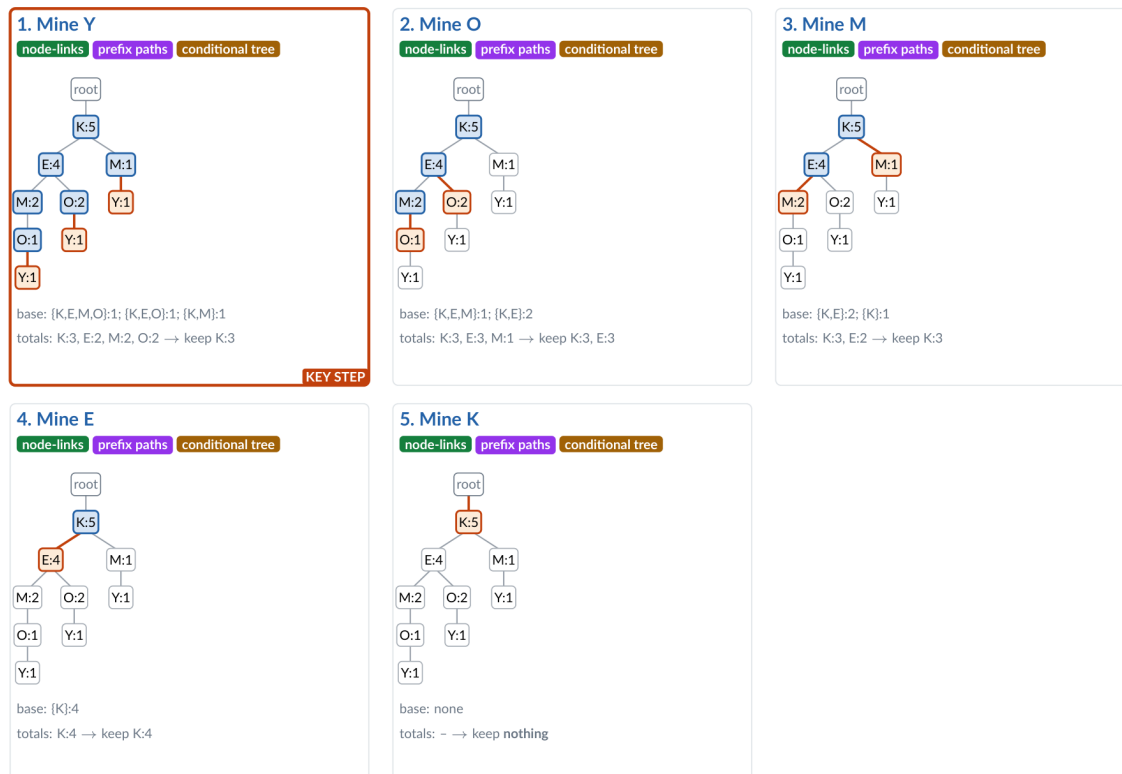
Step 4: mine the tree bottom-up (FP-Growth proper — this is where the marks are)

- The procedure, for one item i at a time, taken in **reverse F-list order** (least frequent first, so $Y \rightarrow O \rightarrow M \rightarrow E \rightarrow K$):
 - Follow i 's node-links to **every** node labelled i .
 - For each such node, write down the **prefix path**: the labels from the root down to it, **excluding i itself**, and tag that whole path with **the count of the i -node**, not the counts written on the path. Together these are i 's **conditional pattern base**.
 - Add up each label's tag totals across the base. Drop labels below min count. What remains is i 's **conditional FP-tree**.
 - Every itemset in that conditional tree, with i appended, is frequent. Its support is the count in the conditional tree.
 - If the conditional tree branches, recurse on it the same way.
- The **single most common mistake** is truncating a prefix path. A path is the *whole* route from the root, so an item sitting between the root and i stays in even when it is not adjacent to i .

Item	Conditional pattern base	Totals $\rightarrow$ cond. FP-tree	Frequent patterns
Y	{K,E,M,O}:1 {K,E,O}:1 {K,M}:1	K:3, E:2, M:2, O:2 $\rightarrow$ only K:3	{K,Y}:3
O	{K,E,M}:1 {K,E}:2	K:3, E:3, M:1 $\rightarrow$ K:3 $\rightarrow$ E:3	{K,O}:3, {E,O}:3, {K,E,O}:3
M	{K,E}:2 {K}:1	K:3, E:2 $\rightarrow$ only K:3	{K,M}:3
E	{K}:4	K:4 $\rightarrow$ K:4	{K,E}:4
K	— (root child, no prefix)	—	{K}:5

- Worked in full for Y**, since it is the one people get wrong. Y sits at the end of three different branches: $K \rightarrow E \rightarrow M \rightarrow O \rightarrow Y:1$, $K \rightarrow E \rightarrow O \rightarrow Y:1$ and $K \rightarrow M \rightarrow Y:1$. Strip the Y and keep **the whole remaining path each time**, so O belongs in the first two entries. Totals give K:3 only, hence the one pattern {K,Y}:3.
- Why only O recurses**: its conditional tree still has two items (K and E), so the single branch $K:3 \rightarrow E:3$ yields {K,O}, {E,O} and the combination {K,E,O}. Every other conditional tree is a single node, so it yields exactly one pattern.

Step by step: mining each item, least frequent first



All frequent itemsets

- 1-itemsets: K:5, E:4, M:3, O:3, Y:3
- 2-itemsets: {K,E}:4, {K,M}:3, {K,O}:3, {E,O}:3, {K,Y}:3
- 3-itemsets: {K,E,O}:3

Strong rules (min conf 80%)

- $K \Rightarrow E = 4/5 = 0.80 \checkmark$ $E \Rightarrow K = 4/4 = 1.0 \checkmark$
- $M \Rightarrow K = 3/3 = 1.0 \checkmark$ $O \Rightarrow K = 3/3 = 1.0 \checkmark$ $Y \Rightarrow K = 3/3 = 1.0 \checkmark$ $O \Rightarrow E = 3/3 = 1.0 \checkmark$
- $K, O \Rightarrow E = 1.0 \checkmark$ $E, O \Rightarrow K = 1.0 \checkmark$ $O \Rightarrow K, E = 1.0 \checkmark$
- Failing, all 7 of them: $K \Rightarrow M$, $K \Rightarrow O$, $K \Rightarrow Y$ and $K \Rightarrow E, O$ (all $3/5 = 0.6$), $E \Rightarrow O$, $K, E \Rightarrow O$ and $E \Rightarrow K, O$ (all $3/4 = 0.75$).
- **Count check:** five 2-itemsets give $5 \times 2 = 10$ rules and {K,E,O} gives $2^3 - 2 = 6$, so **16 candidates = 9 strong + 7 failing.** ✓
- **Note K is on the right of five of the nine.** K is in every transaction, so anything implies it → high confidence, no information (§4.1).

Step by step: the 16 rules against 80%



Problem 3.2 4+4

QUESTION AS PRINTED 2075 Chaitra · Q4 · [4+4]

What do you mean by frequent Pattern growth, draw FP-tree with given tabular data.

TID	Items
01	f, a, c, d, g, i, m, p
02	a, b, c, f, l, m, o
03	b, f, h, j, o, w
04	b, c, k, s, p
05	A, f, c, e, l, p, m, n

Min support is not printed; use 3, which is the standard for this dataset. verified/added

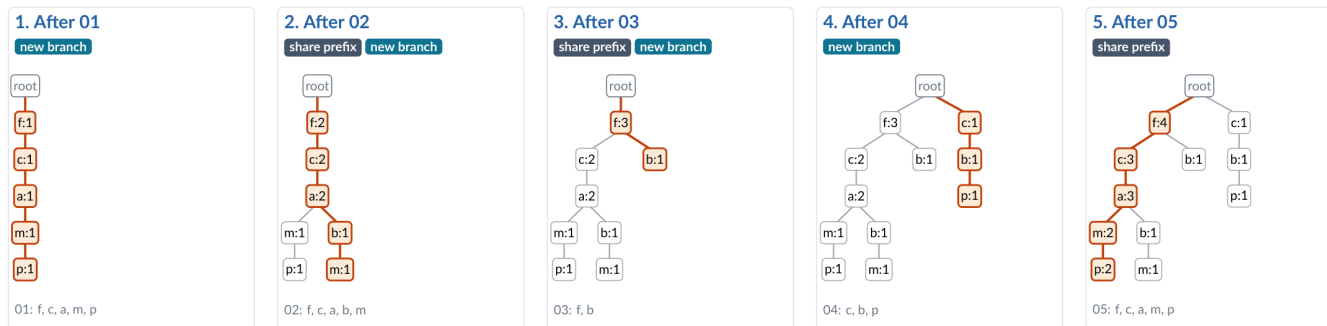
Counts and F-list

- f:4, c:4, a:3, b:3, m:3, p:3 ✓ l:2, o:2, and d,g,i,h,j,k,s,e,n,w all 1 ×
- F-list = f, c, a, b, m, p
- T05 prints a capital "A" — same item as a.

Reordered transactions

- 01 → f, c, a, m, p 02 → f, c, a, b, m
- 03 → f, b 04 → c, b, p
- 05 → f, c, a, m, p

Step by step: the FP-tree after each transaction



The FP-tree

Drawn below, after every transaction; the last panel is the finished tree.

- **Note the second root child:** transaction 04 starts with c, not f, so it cannot share the main prefix.

Problem 3.3 4+4

QUESTION AS PRINTED 2078 Bhadra · Q5 · [4+4]

Consider the given transactional database from a grocery store. Use a support threshold of 33.34% and confidence threshold of 60% to compute the following:

- Build a frequent pattern tree (FP-Tree). Show for each transaction how the tree evolves.
- Use FP-Growth algorithm to discover the frequent itemsets from this FP-tree.

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

- 6 transactions, 33.34% → **min count = 2**.
- Counts: HotDogs 4, Chips 4, Coke 3, Buns 2, Ketchup 2 — **all frequent**.
- F-list = Chips(4), HotDogs(4), Coke(3), Buns(2), Ketchup(2) (4-way tie broken alphabetically).

Tree, built transaction by transaction

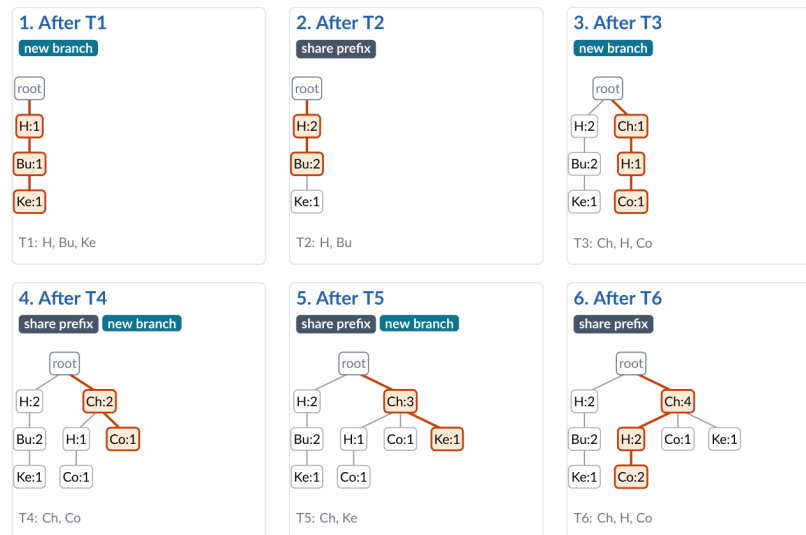
Final tree

Drawn below, after every transaction; the last panel is the finished tree.

T	Reordered	Effect on the tree
T1	H, Bu, Ke	new branch $H:1 \rightarrow Bu:1 \rightarrow Ke:1$
T2	H, Bu	prefix shared $\rightarrow H:2, Bu:2$
T3	Ch, H, Co	new root child $Ch:1 \rightarrow H:1 \rightarrow Co:1$
T4	Ch, Co	$Ch:2$, new child $Co:1$ under Ch
T5	Ch, Ke	$Ch:3$, new child $Ke:1$ under Ch
T6	Ch, H, Co	fully shared $\rightarrow Ch:4, H:2, Co:2$

- **Two root children**, because T1 and T2 do not contain Chips and so cannot join the Chips prefix.

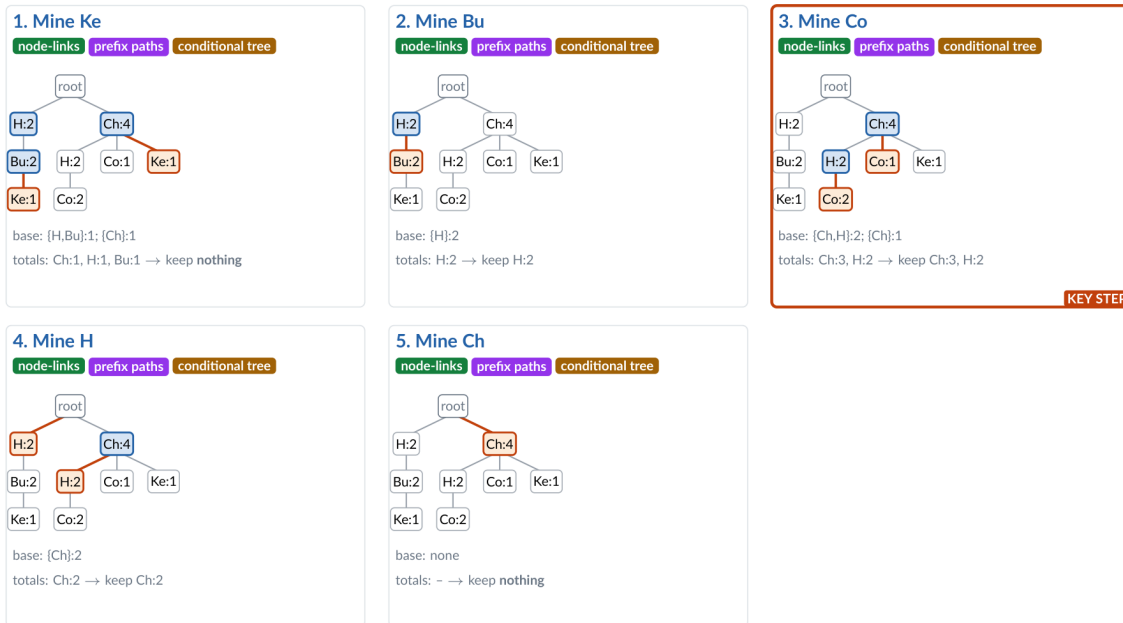
Step by step: the FP-tree after each transaction (orange = the path it touches)



Mining, least-frequent item first

Item	Conditional pattern base	Totals $\rightarrow$ patterns
Ketchup	{H, Bu}:1, {Ch}:1	H:1, Bu:1, Ch:1 $\rightarrow$ all below 2, no pattern
Buns	{H}:2	H:2 $\rightarrow$ {H, Bu}:2
Coke	{Ch, H}:2, {Ch}:1	Ch:3, H:2 $\rightarrow$ {Ch, Co}:3, {H, Co}:2, {Ch, H, Co}:2
HotDogs	{ }:2, {Ch}:2	Ch:2 $\rightarrow$ {Ch, H}:2
Chips	– (root child)	– $\rightarrow$ {Ch}:4

Step by step: mining each item, least frequent first



Frequent itemsets

- **1-itemsets:** Chips 4, HotDogs 4, Coke 3, Buns 2, Ketchup 2
- **2-itemsets:** {Chips,Coke} 3, {Chips,HotDogs} 2, {HotDogs,Coke} 2, {HotDogs,Buns} 2
- **3-itemsets:** {Chips,HotDogs,Coke} 2
- **HotDogs has an empty prefix once** (it is a root child from T1/T2), which contributes nothing to its conditional base. Write the empty set rather than omitting the row.

Problem 3.4 2+8

QUESTION AS PRINTED 2074 Chaitra · Q6 · [2+8]

How does FP growth approach overcomes the disadvantages of Apriori algorithm. For the transaction data given in table generate FP-Tree.

Transaction ID	Item set
T1	Camera, Laptop, Pen drive
T2	Laptop, Pen drive
T3	Laptop, Mobile, Earphone
T4	Earphone, Mobile
T5	Camera, Earphone
T6	Laptop, Mobile, Earphone

Min support is not printed; use 2. Abbreviations: E = Earphone, L = Laptop, Mo = Mobile, Ca = Camera, Pd = Pen drive.

Step 1: counts and F-list

Item	In	Count	Keep
Earphone	T3, T4, T5, T6	4	✓
Laptop	T1, T2, T3, T6	4	✓
Mobile	T3, T4, T6	3	✓
Camera	T1, T5	2	✓
Pen drive	T1, T2	2	✓

F-list = E(4), L(4), Mo(3), Ca(2), Pd(2) – **all five are frequent**, and the E/L tie at 4 is broken alphabetically.

Step 2: reorder each transaction by the F-list

T	As printed	Reordered
T1	Camera, Laptop, Pen drive	L, Ca, Pd
T2	Laptop, Pen drive	L, Pd
T3	Laptop, Mobile, Earphone	E, L, Mo
T4	Earphone, Mobile	E, Mo
T5	Camera, Earphone	E, Ca
T6	Laptop, Mobile, Earphone	E, L, Mo

Step 3: insert them one at a time

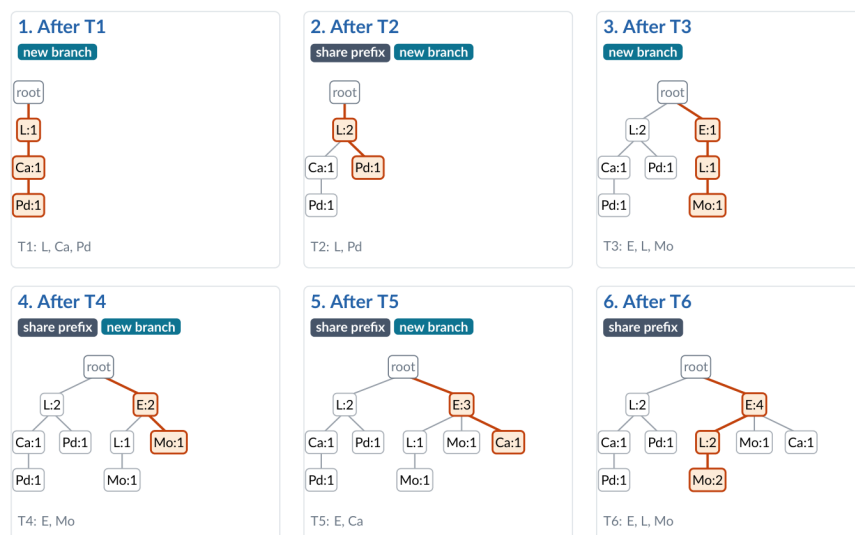
T	Effect on the tree
T1	new branch L:1→Ca:1→Pd:1
T2	L:2, new child Pd:1 under L
T3	new root child E:1→L:1→Mo:1
T4	E:2, new child Mo:1 under E
T5	E:3, new child Ca:1 under E
T6	fully shared → E:4, L:2, Mo:2

- **Two root children:** T1 and T2 hold no Earphone, so they cannot join the E prefix.

Final FP-tree

Drawn below, after every transaction; the last panel is the finished tree.

- **Check:** L appears 4 times in D and the tree shows 2 under the root plus 2 under E. ✓ Do this for one item, it catches most construction errors.
- Mining it gives the key pattern {Earphone, Laptop, Mobile}:2 (T3 and T6). The question only asks for the tree, so this is a bonus line.

Step by step: the FP-tree after each transaction (orange = the path it touches)

M4. Pattern evaluation: lift and correlation

Method

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{conf(A \Rightarrow B)}{sup(B)} \quad \chi^2 = \sum \frac{(obs - exp)^2}{exp}$$

- $lift > 1$ positive · $= 1$ independent · < 1 negative.

Problem 4.1 8

QUESTION AS PRINTED 2070 Chaitra · Q5 · [8]

Why is pattern evaluation important in association rule mining? Explain with example the statistical based measures used for measuring interestingness of association rules.

The paper prints no data — the example is yours to supply. The standard *game* / *video* contingency table below is the one every textbook uses. verified/added

Given

10,000 transactions: 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.

Step 1: the support–confidence view says the rule is strong

- $sup(game \Rightarrow video) = 4000/10000 = 40\%$ (passes a 30% threshold)
- $conf(game \Rightarrow video) = 4000/6000 = 66.7\%$ (passes a 60% threshold)

Step 2: but look at the consequent alone

- $P(video) = 7500/10000 = 75\%$
- **Buying a game DROPS the chance of buying a video from 75% to 66.7%.** The rule is actively misleading.

Step 3: lift catches it

$$lift = \frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89 < 1 \rightarrow \text{negatively correlated}$$

Step 4: χ^2 agrees

	game	<u>game</u>	row
video	4000 (4500)	3500 (3000)	7500
<u>video</u>	2000 (1500)	500 (1000)	2500
col	6000	4000	10,000

$$\chi^2 = \frac{500^2}{4500} + \frac{500^2}{3000} + \frac{500^2}{1500} + \frac{500^2}{1000} = 55.6 + 83.3 + 166.7 + 250 = 555.6$$

- Large $\chi^2 \rightarrow$ correlated; observed (4000) < expected (4500) $\rightarrow$ **negatively**.

Final answer

- **Pattern evaluation matters because support and confidence alone certify rules that are false.** Add a correlation measure before reporting any rule.
- Mention that lift and χ^2 are **not null-invariant**, so for sparse data prefer **all_conf**, **max_conf**, **Kulczynski** or **cosine**.

Step by step: why a 66.7% rule is misleading

1. Support and confidence pass

confidence

sup = 4000/10000 = 40% (over 30%)

conf = 4000/6000 = 66.7% (over 60%)

By these two measures the rule *game* $\Rightarrow$ *video* is strong.

2. Compare with the base rate

confidence

P(video) 75%

P(video|game) 66.7%

Buying a game lowers the chance of a video.

3. Lift

lift

$$\frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89$$

< 1: **negatively** correlated, despite 66.7% confidence.

KEY STEP

4. Chi-square

chi-square

	game	<u>game</u>
video	4000 (4500)	3500 (3000)
<u>video</u>	2000 (1500)	500 (1000)

obs (exp):

$\chi^2 = 55.6 + 83.3 + 166.7 + 250 = 555.6$;

obs < exp for both: negative.

M5. Subsequence enumeration

Method

- A k -**subsequence** contains k items in total (not k elements).
- Choose how many items to take from each element, keeping element order; a chosen element must be non-empty to remain in the sequence.
- **Enumerate systematically** by the split (a, b, c, d) of items taken per element, then remove duplicates.

Problem 5.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (a) · [3]

List all the 4-subsequences contained in the data sequence: $\langle \{1, 3\} \{2\} \{2, 3\} \{4\} \rangle$

Given

$e_1 = \{1, 3\}$, $e_2 = \{2\}$, $e_3 = \{2, 3\}$, $e_4 = \{4\}$; 6 items in total.

What counts as a subsequence (get this right first)

- Take a subset from each element, **possibly empty**. Elements chosen empty simply disappear.
- **Element order must be preserved**, and items from two different elements can never be merged into one element.
- A k -**subsequence** has k items in total, counted across all surviving elements. It is **not** k elements. This is the single most common misreading.
- So $\langle \{1, 3\} \{2\} \rangle$ is a 3-subsequence, not a 2-subsequence.

Why enumerate by splits

- A choice is fully described by **how many items are taken from each element**, written (a, b, c, d) for $e_1 \dots e_4$.
- The bounds are the element sizes: $a \leq 2$, $b \leq 1$, $c \leq 2$, $d \leq 1$, and we need $a + b + c + d = 4$.
- For a fixed split the number of ways is a product of binomials, $\binom{2}{a} \binom{1}{b} \binom{2}{c} \binom{1}{d}$, because the elements are chosen independently.
- **Enumerating splits first guarantees nothing is missed**, which listing by hand does not. Then remove duplicates at the end.

Step 1: enumerate the item-count splits

(a, b, c, d) with $a \leq 2$, $b \leq 1$, $c \leq 2$, $d \leq 1$, sum = 4

(a, b, c, d)	Ways	Subsequences
(2,1,1,0)	2	$\langle \{1, 3\} \{2\} \{2\} \rangle$, $\langle \{1, 3\} \{2\} \{3\} \rangle$
(2,1,0,1)	1	$\langle \{1, 3\} \{2\} \{4\} \rangle$
(2,0,2,0)	1	$\langle \{1, 3\} \{2, 3\} \rangle$
(2,0,1,1)	2	$\langle \{1, 3\} \{2\} \{4\} \rangle$ (duplicate), $\langle \{1, 3\} \{3\} \{4\} \rangle$
(1,1,2,0)	2	$\langle \{1\} \{2\} \{2, 3\} \rangle$, $\langle \{3\} \{2\} \{2, 3\} \rangle$
(1,1,1,1)	4	$\langle \{1\} \{2\} \{2\} \{4\} \rangle$, $\langle \{1\} \{2\} \{3\} \{4\} \rangle$, $\langle \{3\} \{2\} \{2\} \{4\} \rangle$, $\langle \{3\} \{2\} \{3\} \{4\} \rangle$
(1,0,2,1)	2	$\langle \{1\} \{2, 3\} \{4\} \rangle$, $\langle \{3\} \{2, 3\} \{4\} \rangle$
(0,1,2,1)	1	$\langle \{2\} \{2, 3\} \{4\} \rangle$

Step 2: check the count, then remove duplicates

- Ways per split: $2 + 1 + 1 + 2 + 2 + 4 + 2 + 1 = 15$ selections. **Do this arithmetic** — it is how you know no split was dropped.
- **A duplicate appears when the same items can be taken from different elements.** Here $e_2 = \{2\}$ and $e_3 = \{2, 3\}$ share the item 2, so $\langle \{1, 3\} \{2\} \{4\} \rangle$ is produced twice: once as $(2, 1, 0, 1)$ taking the 2 from e_2 , once as $(2, 0, 1, 1)$ taking it from e_3 . **Same sequence, so it counts once.**
- No other pair of splits can collide, because every other split keeps a different number of surviving elements or a different element content.

Final answer

- 15 selections — 1 duplicate $\rightarrow$ **14 distinct 4-subsequences.**
- **The exam usually only wants “at least four”** 79 Bh, so listing the first few from a clearly stated method is enough. **Show the method, not just the list.**

Step by step: enumerate by split, then remove the duplicate

1. Splits (a, b, c, d) summing to 4

item splits

e_1	$\{1, 3\}$	e_2	$\{2\}$	e_3	$\{2, 3\}$	e_4	$\{4\}$	ways
2	1	1	0	2				$\binom{2}{a} \binom{1}{b} \binom{2}{c} \binom{1}{d}$ per split, total 15
2	1	0	1	1				
2	0	2	0	1				
2	0	1	1	2				
1	1	2	0	2				
1	1	1	1	1				
1	0	2	1	2				
0	1	2	1	1				

2. Remove the duplicate

remove duplicates

$\langle \{1, 3\} \{2\} \{4\} \rangle$ arises twice:
 $(2, 1, 0, 1)$: the 2 from e_2 $(2, 0, 1, 1)$: the 2 from e_3
 e_2 and e_3 share the item 2, so the same sequence is reached twice.

$15 - 1 = 14$ distinct 4-subsequences.

KEY STEP

Problem 5.2 3

QUESTION AS PRINTED 2079 Bhadra · Q6 (a) · [3+3]

How do you handle the categorical attributes in data mining process? Explain with example. Generate the at least four subsequences from the given sequence: $\langle \{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\} \rangle$.

- Any order-preserving selection works. Four valid examples:
 - $\langle \{2\} \{6\} \rangle$ $\langle \{3, 5\} \{7\} \{9\} \rangle$
 - $\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$ $\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$
- Must hold:** element order preserved, and each chosen sub-element is a subset of the original element. **Must not do:** reorder elements, or merge items from two elements into one.

Step by step: four order-preserving picks

1. Pick within each element

keep element order

2	3	5	6	7	8	9	1	7	4	$\langle \{2\} \{6\} \rangle$
2	3	5	6	7	8	9	1	7	4	$\langle \{3, 5\} \{7\} \{9\} \rangle$
2	3	5	6	7	8	9	1	7	4	$\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$
2	3	5	6	7	8	9	1	7	4	$\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$

Green = kept. Each pick is a subset of its own element, in element order.

KEY STEP

M6. Subgraph candidate generation by edge growing

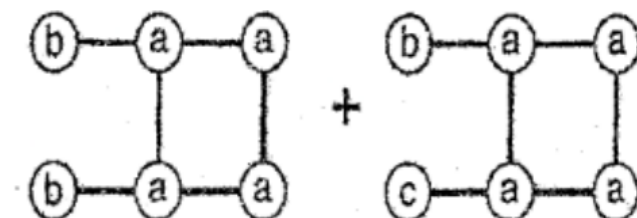
Method

- Two frequent subgraphs are joined only if they share a **common core**: the subgraph left when one edge is removed from each.
- Edge growing** joins two k -edge graphs sharing a $(k - 1)$ -edge core and produces a $(k + 1)$ -**edge candidate** = core + the edge unique to each parent.
- Vertex growing** instead adds a vertex, and generates far more candidates. **Edge growing is preferred for exactly that reason.**
- The trap, and the whole point of this question:** when the core carries **repeated vertex labels**, the two extra edges can be attached in more than one non-isomorphic way, so **one join yields several candidates**. Enumerate them, do not assume there is one.

Problem 6.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (b) · [3]

Draw all candidate sub-graphs obtained from joining the pair of graphs shown below using edge-growing method to expand the sub-graphs.



Step 1: read the two parents

- Both are a **4-cycle of a vertices** with two pendant vertices hanging off two *adjacent* corners of that cycle.
- G_1 : pendants are b and b . G_2 : pendants are b and c .
- Each has **6 vertices and 6 edges**, so $k = 6$.

Step 2: find the core

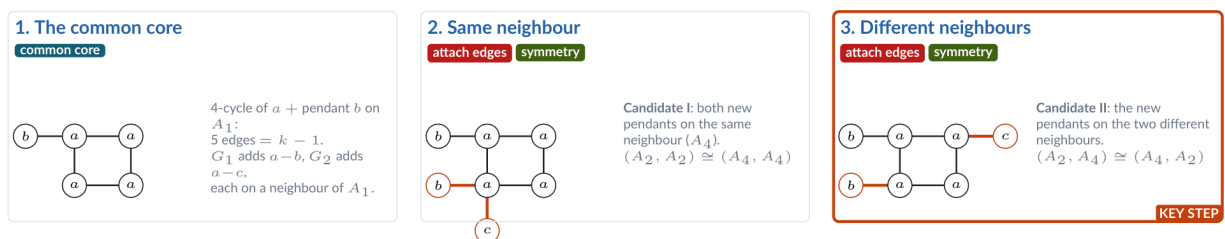
- Delete one edge from each so that what is left is the same graph: drop the second pendant edge from both.
- Core = 4-cycle of a 's + one pendant b** , 5 vertices and 5 edges = $k - 1$. ✓ the graphs are joinable.
- Edge unique to G_1 : $a - b$. Edge unique to G_2 : $a - c$.

Step 3: the candidate size

- Candidate = core + both unique edges $\rightarrow 5 + 2 = 7$ edges, 7 vertices.

Final answer: 2 candidate sub-graphs, each with 7 vertices and 7 edges

Step by step: the core, then the two ways to attach (7 vertices, 7 edges each)



- Marking note:** the marks are for showing the core and then arguing the placements. A single drawing with no core and no count of candidates scores badly even if the drawing itself is one of the two right answers.
- Cross-check:** every candidate must contain both parents as subgraphs, and must have exactly $k + 1 = 7$ edges. Both do.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	k -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
$\rightarrow$	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.